

Rohan Raju Dhanakshirur, PhD

Senior AI Scientist, GE Healthcare | PhD, IIT Delhi

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Senior AI Scientist at GE Healthcare and PhD from IIT Delhi with 8+ years of experience in artificial intelligence, computer vision, and medical AI. Experienced in developing and translating deep learning solutions for healthcare, with expertise spanning explainable and trustworthy AI, medical image analysis, surgical data science, multimodal learning, and video understanding. Author/co-author of 25+ research publications, inventor on three patents, and contributor to award-winning AI systems deployed in clinical and training environments.

Experience

- GE Healthcare** July 2025 – Present
Senior AI Scientist July 2025 – Present
 - Work on explainable and trustworthy AI-based computer vision solutions for ultrasound, focusing on model reliability, failure-mode analysis, and annotation-error detection in medical imaging pipelines.
- Indian Institute of Technology Delhi** August 2018 – July 2025
Senior Project Scientist February 2024 – July 2025
Project Scientist April 2019 – January 2024
Casual Project Staff August 2018 – March 2019
 - Worked on the IITD-AIIMS Collaborative neurosurgery project for the automatic evaluation of endoscopic, microscopic, and neuro-navigation-based neurosurgical skills, from algorithm design through real-world deployment at AIIMS New Delhi.

Education

- Indian Institute of Technology, Delhi** New Delhi, India
PhD - AN and SK School of IT; CGPA: 8.98 August 2019 - July 2026
Thesis: Advancing Automation in Microscopic and Endoscopic Neurosurgical Skills Assessment
Advisors: Prof. Chetan Arora and Prof. Prem Kumar Kalra
- B. V. Bhoomaraddi College of Engineering (VTU-Autonomous)** Hubballi, India
BE - Electronics and Communications; CGPA: 9.84; GOLD MEDAL July 2014 - July 2018

Selected Research Contributions

- AI-Driven Interpretable Evaluation of Micro-Suturing and Micro-Drilling Neurosurgical Skills** (Image-based)
 - Developed a multi-task learning framework for automated and interpretable evaluation of neurosurgical micro-suturing skills from outcome images, with simultaneous prediction of skill-related parameters and overall performance.
 - Work accepted at ICIP 2023 and CIBM 2025; patent filed under the DBT-Indian Patent Cell.
 - System deployed on an NVIDIA Jetson Nano and **being used at AIIMS New Delhi since March 2023.**
- Temporal Action Segmentation and Action Quality Assessment in Surgical Videos** (Video-based)
 - Proposed VideoCutMix, a temporal data augmentation framework for data-scarce surgical video segmentation that preserves optical-flow consistency and generates pseudo action boundaries.
 - Work accepted at MICCAI 2024 (Oral).
 - System deployed on an NVIDIA Jetson AGX and **being used at AIIMS New Delhi since November 2024.**
- Efficient Surgical Instrument Instance Segmentation** (Multimodal)
 - Proposed a learnable query proposal approach to address incorrect query initialization in transformer-based surgical instrument instance segmentation architectures.
 - Developed a multimodal prompt alignment approach for fine-grained classification in foundation-model-based surgical instrument segmentation.
 - Work accepted at MICCAI 2023 (Early Accept).
 - System designed for integration with the video-based surgical skills evaluation system at AIIMS New Delhi.
- Reliability and Annotation-Error Detection in Medical AI** (GE Healthcare)
 - Proposed RELAX (Reliability Enhancement by Limiting Adverse Exertions), a training-time mechanism for improving the reliability of AI systems; accepted at *Nature Scientific Reports*.
 - Proposed MIRROR, a multi-model inference and regional reasoning framework for detecting annotation errors in medical imaging datasets; presented at MIDL 2026.
 - Filed two patents on prompt-driven specialized medical imaging model creation and AI-guided ultrasound acquisition.

Awards and Recognitions

- One of the chief inventors of “Neuromentor”, a system that received the prestigious **Modi-Prix Galien Award 2024** for the Best Academic/Public Sector Innovation.
- Received the **Best Poster Award** for the presentation of the thesis summary at the prestigious **Indian National Academy of Engineers (INAE) Annual Convention 2024**.
- Received the **Best Poster Award** for the presentation of the thesis summary at the **Emerging Technologies for the HealthCare Workshop, AIIMS New Delhi** in October 2024.
- Received the **Best Demonstration Award** for the presentation of the in-house-built neuronavigation system at the **IIT Delhi Open House 2024**.
- Received the **Best Paper Award** at the prestigious **National Systems Conference** in 2024 and 2022.

Skills

- Computer Vision • Medical Image Analysis • Surgical Data Analysis • Deep Learning • Artificial Intelligence
- C/C++ • OOPs • Python • Pytorch • Tensorflow • Git • OpenCV • PIL • SkImage • Translational Research

Selected Publications

- **Rohan Raju Dhanakshirur**, Raghunathan Selvam, Prasad Sudhakar Murthy, Rahul Venkataramani and Chandan Aladahalli. “RELAX: Reliability Enhancement by Limiting Adverse Exertions,” *Nature Scientific Reports*, 2026.
- **Rohan Raju Dhanakshirur**, Keerti KM, Prasad Sudhakar Murthy and Chandan Aladahalli. “MIRROR: Multi-model Inference and Regional Reasoning for Recognizing Annotation Errors,” *MIDL Short Papers*, 2026.
- **Rohan Raju Dhanakshirur**, Mrinal Tyagi, Britty Baby, Ashish Suri, Prem Kumar Kalra, and Chetan Arora. “Video-CutMix: Temporal Segmentation of Surgical Videos in Scarce Data Scenarios,” *27th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024.
- **Rohan Raju Dhanakshirur**, K N Ajay Shastry, Kaustubh Borgavi, Ashish Suri, Prem Kumar Kalra, and Chetan Arora. “Learnable Query Initialization for Surgical Instrument Instance Segmentation,” *26th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023.
- **Rohan Raju Dhanakshirur**, Varidh Katiyar, Ravi Sharma, Ashish Suri, Prem Kumar Kalra, and Chetan Arora. “From Feline Classification to Skills Evaluation: A Multitask Learning Framework for Evaluating Micro Suturing Neurosurgical Skills,” *2023 IEEE International Conference on Image Processing (ICIP)*, pp. 3374–3378.
- **Rohan Raju Dhanakshirur**, Ramesh Ashok Tabib, Ujwala Patil, and Uma Mudenagudi. “Evidence-Based Feature Selection and Collaborative Representation Towards Learning-Based PSF Estimation for Motion Deblurring,” *2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW)*, pp. 1008–1016.
- Rangarajan, Pranjal, Dhruv, **Rohan Raju Dhanakshirur**, Akhil, Chandan, Arun, Smriti Banerjee, and Chetan Arora. “Deep Learning for Detection of Iso-Dense, Obscure Masses in Mammographically Dense Breasts,” *European Radiology*, 2023.
- Britty, Ramandeep, Ashish, **Rohan Raju Dhanakshirur**, Argha Chakraborty, Subodh Kumar, Prem Kumar Kalra, and Subhashis Banerjee. “A Review of Virtual Reality Simulators for Neuroendoscopy,” *Neurosurgical Review*, 2020.

Patents

- Rohan Raju Dhanakshirur, et al. “Microsuturing Simulator with Deep Learning-Based Automated Skills Evaluation,” DBT-Indian Patent (No: 202211071497), Under Process.
- Rohan Raju Dhanakshirur, et al. “Prompt-Driven Creation of Specialized Medical Imaging Models,” GE Healthcare, US Patent Filed.
- Rohan Raju Dhanakshirur, et al. “AI-Guided Ultrasound Acquisition Framework,” GE Healthcare, US Patent Filed.

Guest Lectures

- **Expert Talk** on the “Basics of Image-Processing and Artificial Intelligence” for the Continuing Education Program (CEP) of 50 officials of CRIS in 2024.
- **Expert Talk** on the “Basics of Image Processing” for the IITD-Vision Graphics Summer School in 2024.
- **Organizer and Resource Person** for the project training program to train 20 students from various colleges of the country for one month on the basics of computer vision as a part of the AIIMS-IITD Collaborative Project in 2024.
- **Expert Faculty** for the hands-on session on “Basics of Transformers” as a part of a Workshop on Artificial Intelligence in Modern Biology at ICGB, Delhi, in 2023.
- **Multiple Expert Talks** on topics related to “Research Opportunities in Computer Vision for Medical Community” as a part of the FDP at KLS Gogte Institute of Technology, Belagavi, in 2022–23.
- **Organizer and Resource Person** for the Computer Vision and Graphics workshop conducted at the B. V. Bhoomaraddi College of Engineering and Technology, Hubballi, in Dec-2016, Jun-2017, Dec-2017, Jun-2018, and Dec-2019.
- **Expert Talk** on “Introduction to Machine Learning” for 1st-semester PG students of DKTE Society’s Textile & Engineering Institute in 2018–19.
- **Workshop** on “Introduction to Python” for 1st-semester PG students and 5th-semester UG students of KLE Technological University in 2017–18 as a resource person.